Mermaid Guide

Mermaid Diagram Syntax

> Mermaid's syntax is used to create diagrams. `syntax`, together with `deployment` and `configuration` constitute the whole of mermaid

Diagrams examples can be found in the Mermaid Live Editor

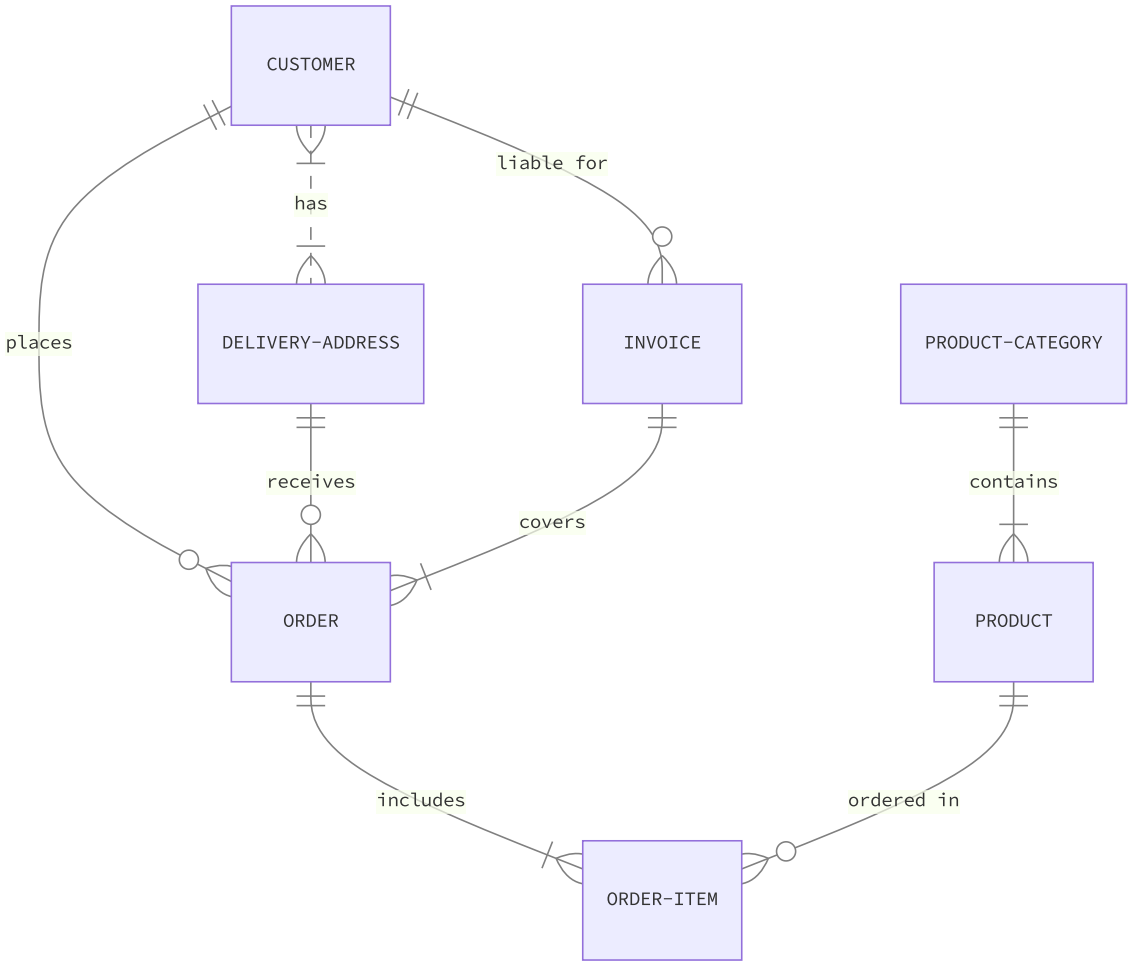
Syntax Structure

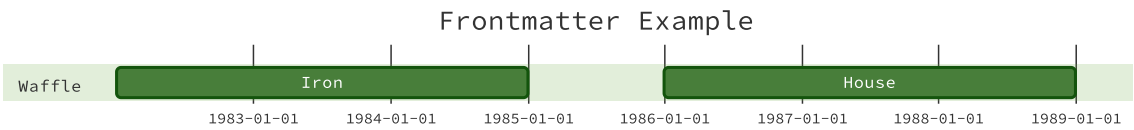
- All `diagrams definitions` begin with a declaration of `diagram type` -> *Notify the parser which kind of diagram to generate*
- Followed by the `definitions` of the diagram and its content

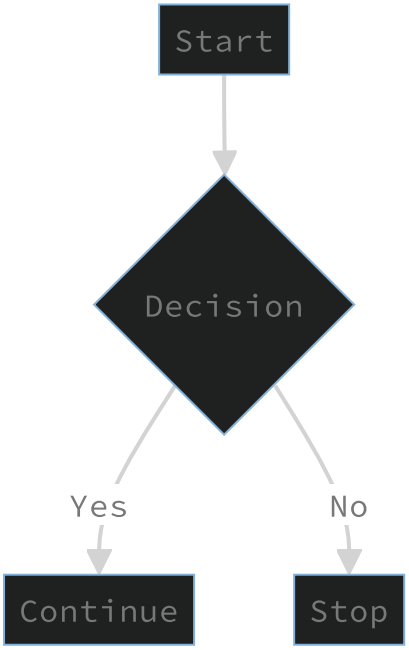
The only exception to this is a Frontmatter configuration

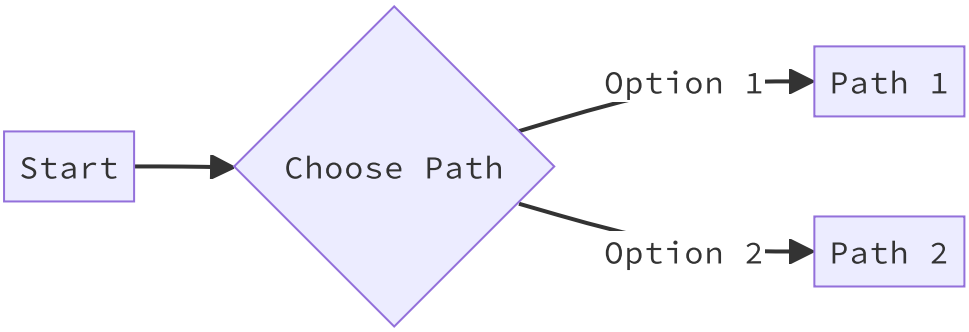
Unknown words and misspellings will break a diagram, while parameters silently fail.

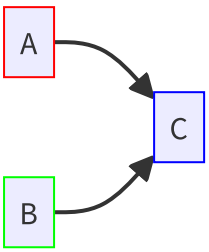
Example: The code below is for an Entity Relationship Diagram, specified by the `erDiagram` declarations. What follows is the definition of different `Entities` represented in it

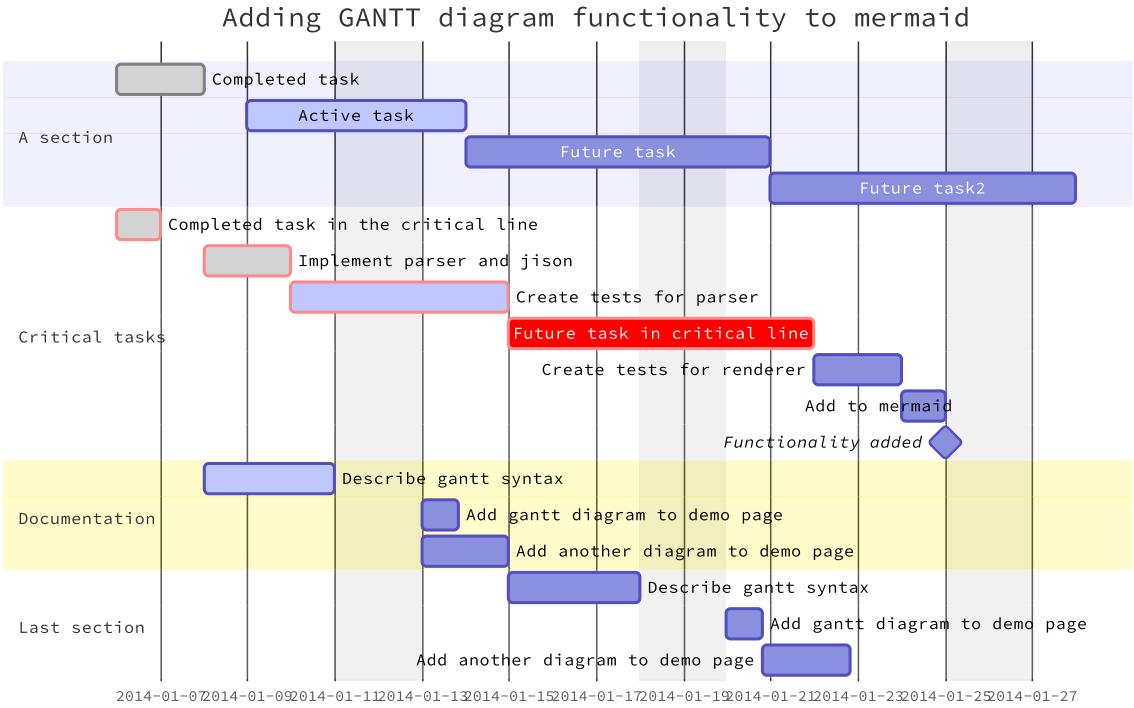

CUSTOMER
has
liable for
places
DELIVERY-ADDRESS
INVOICE
PRODUCT-CATEGORY
receives
covers
contains
ORDER
PRODUCT
includes
ordered in
ORDER-ITEM

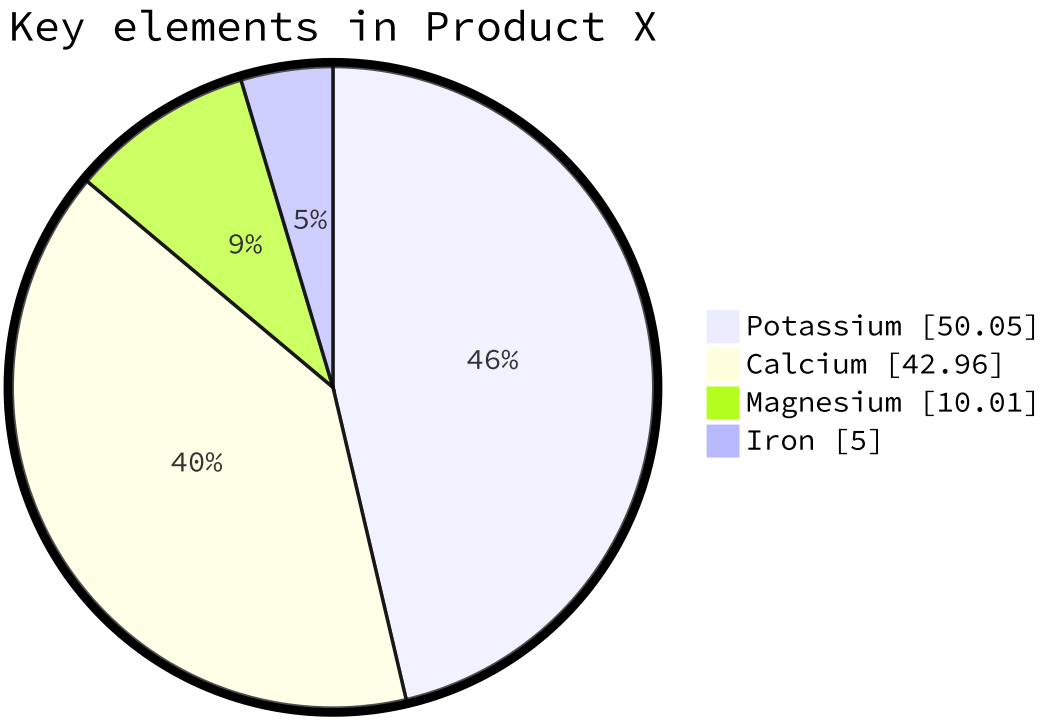

Frontmatter Example
Waffle
Iron
House
1983-01-01
1984-01-01
1985-01-01
1986-01-01
1987-01-01
1988-01-01
1989-01-01

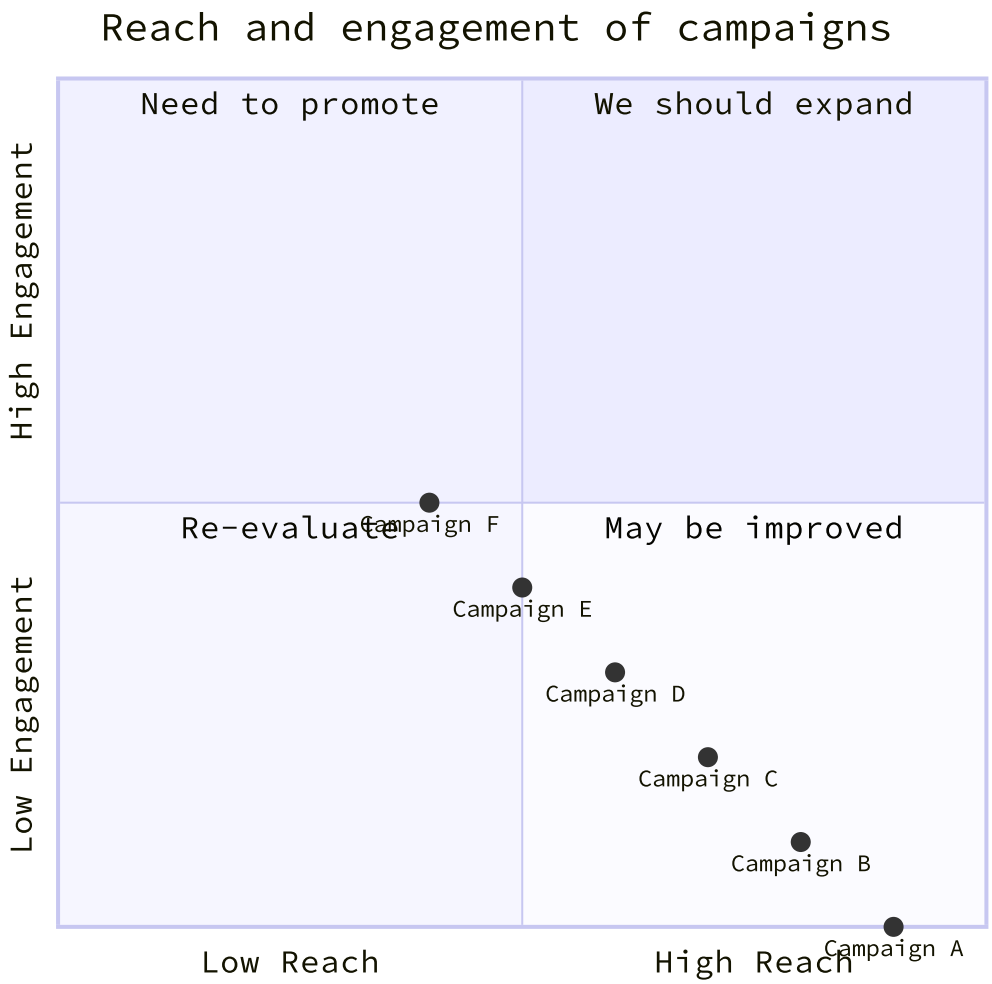

Start
Decision
Yes
No
Continue
Stop

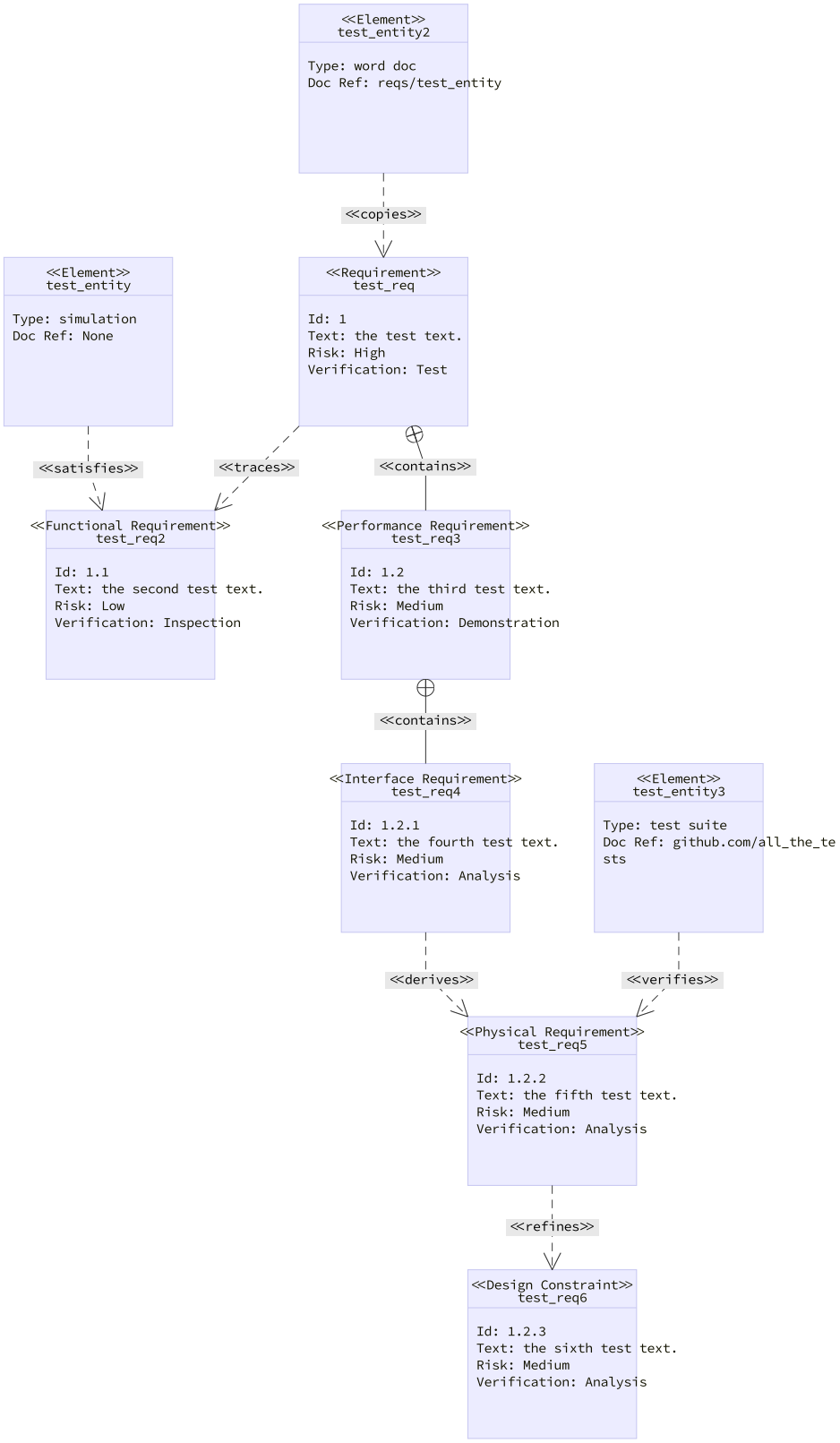

Start
Choose Path
Option 1
Path 1
Option 2
Path 2

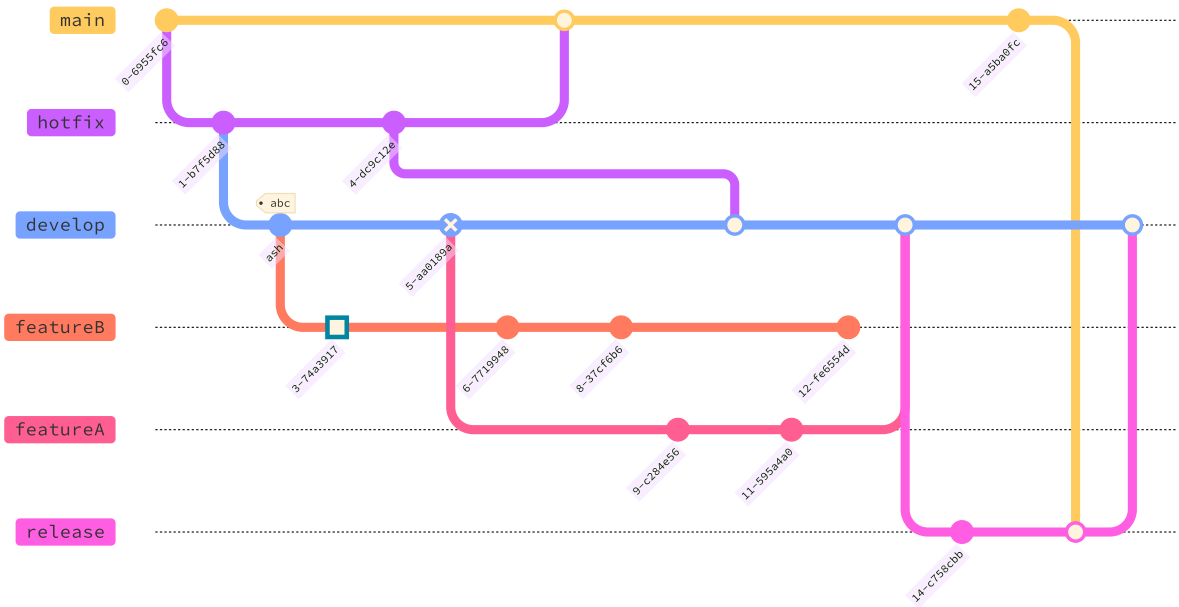

A
B
C

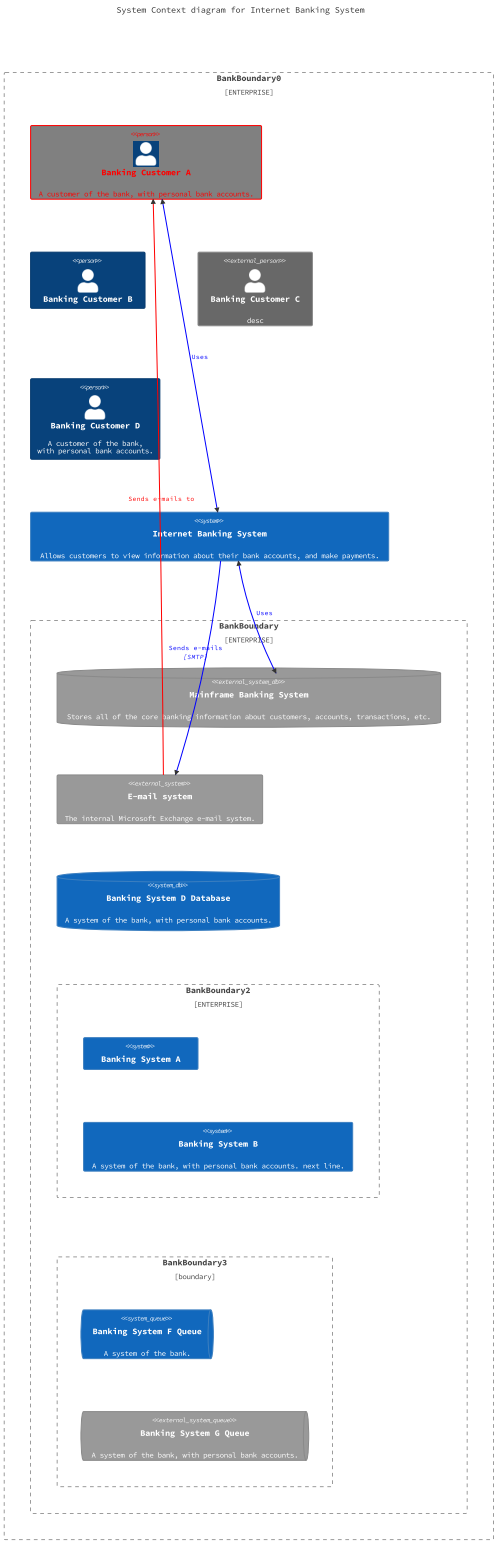

Adding GANTT diagram functionality to mermaid
A section
Completed task
Active task
Future task
Future task2
Critical tasks
Completed task in the critical line
Implement parser and jison
Create tests for parser
Future task in critical line
Create tests for renderer
Add to mermaid
Functionality added
Documentation
Describe gantt syntax
Add gantt diagram to demo page
Add another diagram to demo page
Last section
Describe gantt syntax
Add gantt diagram to demo page
Add another diagram to demo page
2014-01-07
2014-01-09
2014-01-11
2014-01-13
2014-01-15
2014-01-17
2014-01-19
2014-01-21
2014-01-23
2014-01-25
2014-01-27


Key elements in Product X
46%
40%
9%
5%
Potassium [50.05]
Calcium [42.96]
Magnesium [10.01]
Iron [5]


Reach and engagement of campaigns
Need to promote
We should expand
Re-evaluate
May be improved
High Engagement
Low Engagement
Low Reach
High Reach
Campaign F
Campaign E
Campaign D
Campaign C
Campaign B
Campaign A


<<Element>>
test_entity2
Type: word doc
Doc Ref: reqs/test_entity
<<copies>>
<<Element>>
test_entity
Type: simulation
Doc Ref: None
<<Requirement>>
test_req
Id: 1
Text: the test text.
Risk: High
Verification: Test
<<satisfies>>
<<traces>>
<<contains>>
<<Functional Requirement>>
test_req2
Id: 1.1
Text: the second test text.
Risk: Low
Verification: Inspection
<<Performance Requirement>>
test_req3
Id: 1.2
Text: the third test text.
Risk: Medium
Verification: Demonstration
<<contains>>
<<Interface Requirement>>
test_req4
Id: 1.2.1
Text: the fourth test text.
Risk: Medium
Verification: Analysis
<<Element>>
test_entity3
Type: test suite
Doc Ref: github.com/all_the_te
sts
<<derives>>
<<verifies>>
<<Physical Requirement>>
test_req5
Id: 1.2.2
Text: the fifth test text.
Risk: Medium
Verification: Analysis
<<refines>>
<<Design Constraint>>
test_req6
Id: 1.2.3
Text: the sixth test text.
Risk: Medium
Verification: Analysis


main
hotfix
develop
featureB
featureA
release
0-6955fc6
1-b7f5d88
4-dc9c12e
abc
ash
5-aa0189a
3-74a3917
6-7719948
8-37cf6b6
12-fe6554d
9-c284e56
11-595a4a0
14-c758cbb
15-a5ba0fc


System Context diagram for Internet Banking System
BankBoundary0
[ENTERPRISE]
<<person>>
Banking Customer A
A customer of the bank, with personal bank accounts.
<<person>>
Banking Customer B
<<external_person>>
Banking Customer C
desc
Uses
<<person>>
Banking Customer D
A customer of the bank, with personal bank accounts.
Sends e-mails to
<<system>>
Internet Banking System
Allows customers to view information about their bank accounts, and make payments.
Uses
BankBoundary
[ENTERPRISE]
Sends e-mails (SMTP)
<<external_system_db>>
Mainframe Banking System
Stores all of the core banking information about customers, accounts, transactions, etc.
<<external_system>>
E-mail system
The internal Microsoft Exchange e-mail system.
<<system_db>>
Banking System D Database
A system of the bank, with personal bank accounts.
BankBoundary2
[ENTERPRISE]
<<system>>
Banking System A
<<system>>
Banking System B
A system of the bank, with personal bank accounts. next line.
BankBoundary3
[boundary]
<<system_queue>>
Banking System F Queue
A system of the bank.
<<external_system_queue>>
Banking System G Queue
A system of the bank, with personal bank accounts.

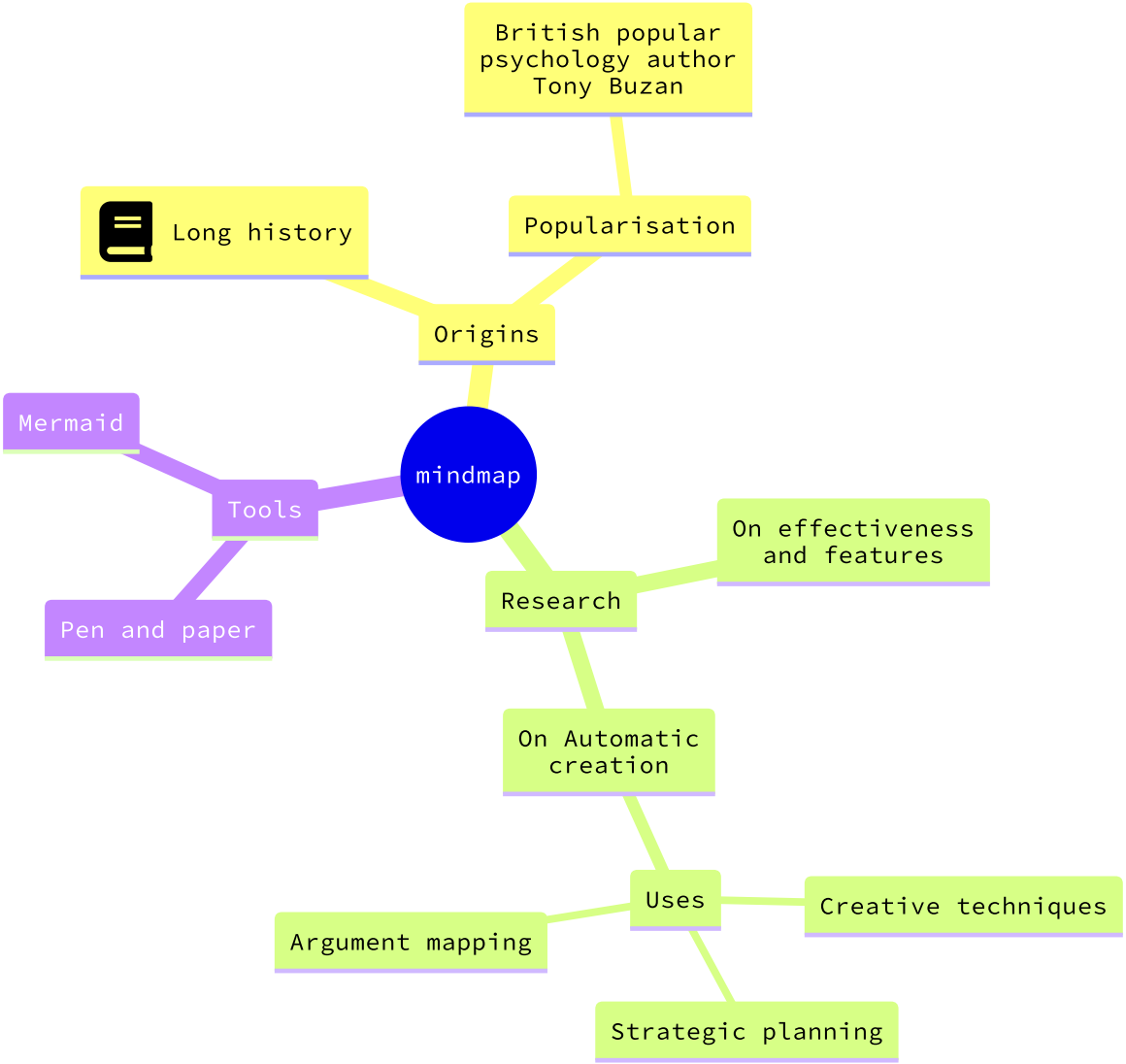


Diagram Breaking

One should beware that the use of some words or symbols that can break diagrams. These words or symbols are few and often only effect specific types of diagram.

Diagram Breakers	Reason	Solution
comments	N A	N A
%%(“)%%	Similar to Directive confuses the renderer	In comments using %%, avoid using ”{}”
flow-charts	N A	N A
'end'	The word ”End” can cause flowcharts and Sequence diagrams to break	wrap them in quotation marks to prevent breakage
Nodes inside Nodes	Mermaid gets confused with nested shapes	wrap them quotation marks to prevent breaking

Frontmatter of Diagram Code